ID Y08-MA-W01-AK rev 2b8ceb

Integers, Primes & Powers — the building blocks of number

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Worksheet

Part A

1. $\frac{9}{12} + \frac{8}{12} = \frac{17}{12} = 1\frac{5}{12}$
2. 10% = 24, 5% = 12 → **36**
3. $6 \times 7 = 42$, 3 decimal places → **0.042**
4. $\frac{5}{2} \times \frac{6}{5} = \frac{30}{10} =$ **3**
5. $48 \times 25 = 12 \times 100 =$ **1200** (since $4 \times 25 = 100$)
6. **3.14**

Diagnostic: errors on 1, 3 or 4 suggest fraction/decimal revision is needed before Week 3.

C1

1. a) −8 b) 8 c) −56 d) 8 e) −27 f) −9
g) $5 - (-12) = 17$
h) $(-8)(-5) + 4 = 40 + 4 = 44$
i) $-24 \div (-6) - 9 = 4 - 9 = -5$
j) $[18 - 3(-5)] \div (-11) = 33 \div (-11) = -3$

2. a) $2^3 \times 3^2$ b) $2 \times 3 \times 5^2$ c) $2^2 \times 7^2$ d) $7 \times 11 \times 13$ e) $2^3 \times 11 \times 23$ ($2024 \div 8 = 253 = 11 \times 23$) f) $2^4 \times 3^2 \times 5^2$

3. a) 32 b) 81 c) 1 000 000 d) 13 e) $2^2 \times 3 = 12$ f) 7 g) $64 - 81 = -17$

C2

4.

Numbers	Prime factorisations	HCF	LCM
36, 60	$2^2 \cdot 3^2$; $2^2 \cdot 3 \cdot 5$	12	180
45, 98	$3^2 \cdot 5$; $2 \cdot 7^2$	1	4410
24, 40, 90	$2^3 \cdot 3$; $2^3 \cdot 5$; $2 \cdot 3^2 \cdot 5$	2	360

45 and 98 share no prime factors — they are **coprime** (relatively prime), so HCF = 1 and LCM = their product.

5. a) $(6 + 4) \times 3 - 1 = 29$ b) $(6 + 4) \times (3 - 1) = 20$ c) $6 + 4 \times (3 - 1) = 14$ d) $-2 \times (5 - 8) \div 2 = 3$

6.
a) **True.** Product of two negatives is positive; their sum is negative; a positive is always greater than a negative.

b) **False.** 2 is prime and even (the only even prime).
c) **True.** Pairs of (-1) multiply to $+1$; an odd number of factors leaves one (-1) unpaired.
d) **False.** e.g. LCM(4, 6) = 12, not 24. True only when the numbers are coprime.

7. Sam subtracted first $(20 - 12)$. Division and multiplication come before subtraction: $12 \div 4 = 3$, $3 \times (-3) = -9$, $20 - (-9) =$ **29**.

8. a) 8 hours × 3 = +24 → **−3 °C** b) 7 hours × (−5) = −35 → **−38 °C** c) $-3 - (-38) =$ **35 °C**

C3

9.
a) $1764 = 2^2 \times 3^2 \times 7^2$ → all indices even → **perfect square**, $\sqrt{1764} = 2 \times 3 \times 7 = 42$. Not a cube.
b) $5832 = 2^3 \times 3^6$ → indices multiples of 3 → **perfect cube**, $\sqrt[3]{5832} = 2 \times 3^2 = 18$. Not a square (index of 2 is odd).
c) $2700 = 2^2 \times 3^3 \times 5^2$ → **neither**.

10. a) Multiply by **3** → $2^2 \cdot 3^4 \cdot 5^2 = 8100 = 90^2$. b) Multiply by $2 \times 5 =$ **10** → $2^3 \cdot 3^3 \cdot 5^3 = 27\,000 = 30^3$.

11. $168 = 2^3 \times 3 \times 7$. Need every index even → $n = 2 \times 3 \times 7 =$ **42**. $168 \times 42 = 7056 = 84^2$, so $\sqrt{168n} = 84$.

C4 Connections

X1. $1764 = 2^2 \cdot 3^2 \cdot 7^2$ → side $= 2 \cdot 3 \cdot 7 = 42$ m; perimeter $= 4 \times 42 = 168$ m.
X2. $5832 = 2^3 \cdot 3^6$ → edge $= 2 \cdot 3^2 = 18$ cm; surface area $= 6 \times 18^2 = 6 \times 324 = 1944$ cm².
X3. Width $= \dfrac{2^5 \times 3^2}{2^4} = 2 \times 3^2 = 18$ cm (previews the quotient law, Week 2); perimeter $= 2(16 + 18) = 68$ cm.
X4. HCF(84, 120) = 12 → $\dfrac{7}{10}$.
X5. a) LCM 36: $\dfrac{15}{36} + \dfrac{14}{36} = \dfrac{29}{36}$ b) LCM 120: $\dfrac{5}{120} - \dfrac{3}{120} = \dfrac{2}{120} = \dfrac{1}{60}$
X6. a) $9 - 2(-3)(4) = 9 + 24 = 33$ b) $(-7)^3 = -343$ c) $\dfrac{16 - (-3)}{1} = 19$

Part D

12. a) LCM(12, 18) = 36 → **7:36 am**. b) 7:00, 7:36, 8:12, 8:48, 9:24, 10:00 → **6 times**.

13. Work in cm: 420 and 360. HCF(420, 360) = 60 ($420 = 2^2 \cdot 3 \cdot 5 \cdot 7$, $360 = 2^3 \cdot 3^2 \cdot 5$). Largest tile **60 cm × 60 cm**; $7 \times 6 =$ **42 tiles**.
Common error: using 4.2 and 3.6 directly — highlight converting to whole units first.

14. HCF(84, 126, 210) = $2 \times 3 \times 7 = 42$ → **42 bags**, each with **2 muesli bars, 3 juice boxes, 5 apples**.

15. a) Let c = correct. $4c - 2(25 - c) = 46 \Rightarrow 6c - 50 = 46 \Rightarrow c = 16$. (Guess-and-check or a table is fine at this stage: 16 × 4 − 9 × 2 = 64 − 18 = 46 ✔.)
b) **No.** Every score is $4 \times (\text{correct}) - 2 \times (\text{incorrect})$ — both terms are even, so every possible score is even. 47 is odd.

Part E

16. a) 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.
b) A factor is $2^i \times 3^j$ where i can be any of $0..a$ ($a + 1$ choices) and j any of $0..b$ ($b + 1$ choices); every pair of choices gives a different factor → $(a + 1)(b + 1)$.
c) **60** ($2^2 \cdot 3 \cdot 5$ → $3 \times 2 \times 2 = 12$). Check smaller candidates: 48 has 10, 36 has 9; others below 60 have fewer.

17. Last digits of powers of 7 cycle: 7, 9, 3, 1 (period 4). $2026 = 4 \times 506 + 2$ → same as 7^2 → **9**.

18. Among any three consecutive integers, at least one is even (so the product has a factor 2) and exactly one is a multiple of 3 (so the product has a factor 3). Since 2 and 3 are coprime, the product is divisible by $2 \times 3 = 6$.

Homework

1. a) $-15 + 7 + 3 = -5$ b) $16 - 16 = 0$ c) $(-8) \times (-2) = 16$ (left to right) d) $-3 \times 6^2 = -108$ e) $-1 + 1 = 0$

2. a) $252 = 2^2 \times 3^2 \times 7$ b) $945 = 3^3 \times 5 \times 7$ c) $1024 = 2^{10}$

3. HCF $= 3^2 \times 7 = 63$; LCM $= 2^2 \times 3^3 \times 5 \times 7 = 3780$. Check: $63 \times 3780 = 238\,140 = 252 \times 945$ ✔

4. LCM(15, 20, 25) = $2^2 \times 3 \times 5^2 = 300$ s = **5 minutes**.

5. LCM(2, 3, 4, 5, 6) = 60, so the number is $60k + 1$: 61, 121, 181, 241, **301**. $301 = 7 \times 43$ ✔ → **301**.

6. a) $\frac{6}{20} = \frac{3}{10}$ b) $\frac{21}{12} - \frac{10}{12} = \frac{11}{12}$ c) $92 d) $\frac{3}{8}$

7. Continuation: 3, 6, 9. Each line the multiplier goes down by 1 and the answer goes **up** by 3; to keep the pattern going, $-3 \times (-1)$ must be $+3$. Accept any clear explanation (also: number-line / "removing debts" models).

Teacher notes

- **Common misconceptions:** -3^2 vs $(-3)^2$; doing ÷/× strictly in the order "D then M" rather than left to right; thinking 1 is prime; LCM = product always.
- **If the student finishes early:** ask them to *prove* HCF × LCM = product for two numbers using the "lowest power / highest power" rule (each prime's two indices get used exactly once between the HCF and LCM).
- **If the student struggles:** do factor trees side-by-side and use a Venn diagram of prime factors for HCF/LCM (shared primes in the overlap = HCF; whole diagram = LCM).
- **Facts:** bus frequencies and the Antarctic temperatures are hypothetical (the question says "suppose").
- **Next week builds on:** index notation → index laws; prime factorisation is reused for simplifying surds-style square roots and estimating roots.

Fast-finisher bank — answers

F1. a) $-32 + 27 = -5$ b) $(36 + 24) \div 4 = 15$ c) $-16 + 16 = 0$ d) $3 - 4 \times 13 \div 13 = 3 - 4 = -1$

F2. $9450 = 2 \times 3^3 \times 5^2 \times 7$; $13\,860 = 2^2 \times 3^2 \times 5 \times 7 \times 11$.
HCF $= 2 \times 3^2 \times 5 \times 7 = 630$; LCM $= 2^2 \times 3^3 \times 5^2 \times 7 \times 11 = 207\,900$.
Check: $630 \times 207\,900 = 130\,977\,000 = 9450 \times 13\,860$ ✔

F3. $12 \times 360 \div 72 = 60$. HCF(72, 60) = 12 ✔, LCM = 360 ✔.

F4. Write the numbers as $6m$ and $6n$ with m, n sharing no factors and $6mn = 72 \Rightarrow mn = 12$. Coprime pairs: (1, 12), (3, 4) [(2, 6) fails — HCF would be 12]. **(6, 72) and (18, 24).**

F5. LCM(24, 36) = 72 teeth must pass → small gear **3 turns**, large gear **2 turns**.

F6. LCM = $60 = 2^2 \times 3 \times 5$; make every index even → $2^2 \times 3^2 \times 5^2 =$ **900**.

F7. Count factors of 5 (there are plenty of 2s): multiples of 5 up to 25 give 5, and 25 gives one extra → **6 zeros**.

F8. $6^4 \times 12 = 2^4 3^4 \times 2^2 3 = 2^6 \times 3^5 \rightarrow a = 6, b = 5$.

F9. a) $15 \times 13 \times 6 = 1170$. b) Expanding gives every product $2^i 3^j 5^k$ exactly once — i.e. every factor of 360 exactly once — so the total is the sum of all factors. c) $(1 + 2 + 4 + 8)(1 + 3 + 9) = 15 \times 13 = 195$ (check: 1+2+3+4+6+8+9+12+18+24+36+72 = 195 ✔).

F10. a) Lockers **1, 4, 9, 16, 25, 36, 49, 64, 81, 100** (the perfect squares). b) Locker n is changed once for each factor of n. Factors come in pairs ($a \times b = n$), so most numbers have an even number of factors → ends closed. Only perfect squares have an unpaired factor ($\sqrt{n} \times \sqrt{n}$) → odd number of changes → open. Links to Q16 and Part B §5.

F11. a) e.g. 4 = 2+2, 6 = 3+3, 8 = 3+5, 10 = 3+7, 12 = 5+7, 14 = 7+7, 16 = 3+13, 18 = 5+13, 20 = 3+17, 22 = 3+19, 24 = 5+19, 26 = 3+23, 28 = 5+23, 30 = 7+23.
b) **4 ways:** 3+47, 7+43, 13+37, 19+31.
c) Odd = odd + even, and the only even prime is 2, so an odd number n is a sum of two primes only if $n - 2$ is prime. $27 - 2 = 25$ is not prime, so 27 can't be done.